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RESEARCH ARTICLE

# A Low-Cost AI-Powered System for Early Detection of Diabetic Retinopathy and Ocular Diseases in Resource Limited Settings

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**ABSTRACT** Eye Health is a vital yet often neglected aspect of healthcare, especially in distant and deprived regions that are less accessible to specialized diagnostic tools and professionals. This paper introduces an intelligent diabetic retinopathy detection system to enhance affordability, accessibility, and early diagnosis. The system features a cost-effective imaging setup utilizing a 28D lens and a camera to capture high-quality fundus images. These images are analyzed using advanced Convolutional Neural Networks to determine the cases of people suffering from eye issues: Age-related vascular degeneration (AMD), Diabetic retinopathy (DR), and Glaucoma. A user-friendly web application hosted on AWS facilitates seamless image upload, automated diagnosis, and report generation. Achieving an accuracy of 96.5%, the system ensures reliable diagnostics, enabling timely medical intervention. By integrating AI-driven analysis and cloud-based accessibility, this solution aims to transform eye care in resource-limited settings. Performance evaluation against existing methods highlights its superiority, with an F1 score of 98.2 % for DR, 92.5% for Glaucoma, and 98.8% for AMD, outperforming many previous approaches that either focused on single–disease detection or relied on costly hardware.

**INDEX TERMS** Artificial intelligence, convolutional neural network, diabetic retinopathy, eye disease.

## I. INTRODUCTION

Vision is one of the most vital senses, enabling individuals to navigate and interpret the world. However, despite its critical importance, eye health care remains severely neglected in many regions, particularly in rural and underserved areas [1]. The barriers to effective eye care are multifaceted, including the exorbitant cost of diagnostic equipment, a shortage of trained ophthalmologists, and the time-consuming nature of conventional diagnostic processes.

Several debilitating eye problems, including Diabetic Retinopathy (DR) [2], [3], [4], Glaucoma [5], [6], [7], and Age-related Macular Degeneration (AMD) [8], [9], [10], often progress without noticeable symptoms in their early stages, leading to permanent vision loss if not diagnosed and treated well in time. A major complication of diabetes, DR affects the retina's blood vessels, causing mild symptoms such as occasional vision fluctuations or difficulty reading in its early stages. If left unchecked, it progresses to more severe manifestations like dark spots in vision, blurred or distorted sight, and, in extreme cases, retinal detachment or macular edema, which can result in blindness. Glaucoma, known as the "silent thief of sight" [5], [6], [7], causes gradual peripheral vision loss and tunnel vision in its advanced stages. Patients may also experience blurred vision, halos around lights, and difficulty adapting to low-light

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conditions. A common risk factor is increased intraocular pressure; normal-tension glaucoma can still occur without elevated pressure. If untreated, glaucoma leads to irreversible blindness [11]. AMD primarily affects central vision, impairing daily reading, driving, and facial recognition. Initial symptoms include trouble seeing at night and blurry central vision.

Straight lines may appear wavy as they progress, and complete central vision loss can occur. The advanced wet form of AMD involves abnormal blood vessel growth and leakage under the retina, causing sudden and severe visual deterioration. Early detection is critical to slowing disease progression.

Traditional diagnostic methods heavily rely on expensive fundus cameras [12] and highly specialized medical professionals, often unavailable in rural or low-resource settings. While artificial intelligence [4] (AI)-driven diagnostic tools have recently emerged as a promising alternative, these systems have significant limitations. Many AI-based solutions depend on costly imaging devices and computationally expensive cloud-based models [12], making them impractical for widespread deployment in underserved areas. Most AI solutions are designed to detect only one specific disease, necessitating multiple diagnostic tools for comprehensive eye care. Many existing platforms lack user-friendly interfaces, making it difficult for non-specialists or local healthcare workers to utilize them effectively. Moreover, variability in the datasets used to train these models can lead to inconsistent performance, making them less reliable across diverse populations. Some solutions even fail to communicate results seamlessly to patients and caregivers. These barriers often delay diagnosis and treatment, increasing the likelihood of permanent vision impairment, which can severely hinder a person's capability to work and be involved in everyday activities. A practical solution would combine cost-effective hardware and reliable machine-learning algorithms to provide accurate diagnostics [30] without relying on expensive equipment or highly specialized expertise. Developing tools that can work effectively with minimal infrastructure while maintaining high diagnostic performance across multiple diseases is important. Addressing these gaps could help improve the early detection of eye diseases, enabling timely treatment and potentially preventing vision loss. Although this is not a one-size-fits-all solution, a well-designed system could significantly contribute to reducing the global burden of preventable blindness and improving access to quality eye care, particularly for vulnerable populations in underserved regions.

### A. MOTIVATION

Addressing the critical barriers of cost, accessibility, and usability, this system paves the way for democratizing eye care services. Detecting vision problems early and providing timely treatment can greatly lessen the impact of vision loss, enhancing the quality of life for millions across the globe. This holistic approach—leveraging cost-effective hardware, cutting-edge AI, and scalable cloud technology—significantly advances global eye health care, particularly for underserved communities. A comparative analysis of the most recent papers is shown in Table 1. It highlights the differences between the different works based on various parameters like use of the ensemble model, multiple disease detection, use of transfer learning, cost-effectiveness, etc. Unlike prior works that require high-end fundus cameras or heavyweight cloud CNNs, our design unifies an ESP32-CAM capture block and a 3.5 M-parameter custom CNN so that end-to-end inference runs on a low-cost microcontroller or on AWS as a fallback, cutting hardware cost by ≈70% versus commercial systems. This integrated edge–cloud dual-mode deployment is absent from earlier multi-disease studies that host only in the cloud and thus remain unusable offline.

### B. CONTRIBUTION

This research introduces a novel solution designed to tackle the challenges of diagnosing eye diseases in low-resource settings by leveraging advancements in affordable hardware and machine learning. The focus is on diagnosing the three major diseases using just one system. The main contributions of this study are outlined as follows:

i) Developing a novel integrated eye disease detection system incorporating cost-effective components, including a 28D lens and an ESP32 microcontroller. These low-cost, portable components significantly reduce the overall diagnostic cost, making the solution affordable for healthcare providers in underserved and rural areas. A high-accuracy convolutional neural network (CNN) identifies and diagnoses multiple eye diseases: DR, AMD, and glaucoma. This capability allows the system to address co-existing conditions in a single diagnostic process, enhancing its clinical utility.

ii) Real-Time Cloud-Based Analysis: The proposed solution uses cloud-based processing to analyze real-time images. This ensures that diagnostic feedback is provided promptly, enabling early detection and timely intervention. By offloading computational tasks to the cloud, the system can operate on lightweight hardware, further enhancing its portability and affordability.

iii) Diagnostic Performance: The proposed system demonstrates a high overall (average) diagnostic accuracy of 95.5%, comparable to, or even exceeding, traditional diagnostic methods in certain scenarios. The system incorporates advanced data augmentation and preprocessing techniques to improve generalization. This reduces the risk of overfitting and ensures consistent performance even when applied to datasets with significant variability. The design prioritizes robustness and adaptability, enabling the system to perform well across diverse patient demographics and imaging conditions.

iv) A user-friendly Interface: A web application has been developed to integrate seamlessly with the diagnostic tool. This interface simplifies the process for healthcare providers,

**TABLE 1. Comparison of previous papers based on different parameters. EM – Ensemble Models, MDD- multiple disease detection, TL- Transfer Learning, IFA- Image focus assessment, OT- Optimization Techniques, CE- Cost Effectiveness, CDU- Clinical Data Usage, ED- Early Detection, Acc- Accuracy.**

| Ref | Authors | Year | EM | MDD | TL | IFA | OT | CE | CDU | ED | Acc (%) | Cited |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| [23] | Gauransh Sawhney,Daksh Dave | 2023 | √ | × | √ | √ | √ | × | √ | √ | 93 | - |
| [24] | Israa Odeh, Mouhammd Alkasassbeh | 2021 | √ | × | √ | √ | √ | × | √ | √ | 95.6 | 35 |
| [25] | Mahesh B S, Priya S | 2024 | √ | × | √ | √ | √ | × | √ | √ | 98.09 | - |
| [14] | A A. A, El-Khalek at al. | 2024 | √ | × | √ | √ | √ | × | √ | √ | 96.85 | 4 |
| [26] | Xiangjie Leng, Ruijie Shi | 2023 | √ | × | √ | √ | √ | × | √ | √ | 94.5 | 23 |
| [27] | Najeeba Afreen, Rajanikanth Aluvalu | 2024 | √ | × | × | √ | √ | × | √ | √ | 94.7 | 2 |
| [19] | Qiaowei Wu, Yijun Hu | 2022 | √ | × | √ | × | √ | × | √ | √ | 96.5 | 30 |
| [20] | Isaac Arias-Serrano, Paolo A. Velásquez-López | 2024 | × | × | √ | √ | √ | √ | √ | √ | 94.3 | 27 |
| [21] | Nasser Tamim, Mohamed Elshrkawey | 2021 | × | × | × | √ | √ | √ | √ | √ | 96.67 | 20 |
| [22] | Ruoan Han, Gangwei Cheng | 2022 | × | √ | √ | √ | √ | √ | √ | √ | 95 | 11 |
| | Proposed technique | - | | √ | √ | √ | √ | √ | √ | √ | 96.5 | |

enabling them to upload images, receive diagnostic results, and interpret findings efficiently. The ease of use reduces the time required for diagnosis and expedites treatment decisions.

v) Portable, scalable, and Affordable: The system is designed to be compact, lightweight, and of low cost to allow its use in remote and resource–limited environments. The easy accessibility makes it useful to a broad range of users, including primary care providers in rural areas, bridging the gap between technological advances and practical healthcare needs and contributing to better public health outcomes.

### C. ARTICLE ORGANIZATION

The paper is organized as follows: Section II offers an extensive review of the related literature, highlighting international and national research. Section III elaborates on the proposed methodology in detail. Section IV describes the experimental setup and examines various test scenarios used to. Evaluate the experimental setup and examine various test scenarios used to evaluate the experimental setup used to evaluate the effectiveness of the system. Lastly, section V summarizes the key findings and provides recommendations for future research directions.

## II. LITERATURE REVIEW

For diabetic retinopathy, deep learning architectures, such as convolutional Neural Networks (CNN) [12], [13] have proven effective in identifying and recognizing retinal abnormalities with high accuracy, achieving diagnostic accuracies of up to 93.6% in some studies. DR affects approximately 30% of people with diabetes globally. However, challenges such as dataset variability, inconsistent image quality, and limited clinical validation reduce the generalizability of these models. Machine learning models have often achieved sensitivities and specificities exceeding 90% for glaucoma. Despite these advancements, the reliance on expensive imaging equipment and AI systems' ''opaque model'' nature hinders their widespread adoption. Glaucoma, if left untreated, causes irreversible blindness in approximately 10% of cases. In the case of AMD, deep learning models have shown promise, with accuracies exceeding 90% [14] for detecting early and advanced stages of the disease. However, these solutions' lack of diverse datasets and limited scalability prevents their widespread implementation, especially in underserved areas. A comparative review of prior research is presented in Table 1. The deep learning model proposed by Fong [2] has significantly enhanced the detection and diagnosis of age-related macular degeneration (RD). Convolutional Neural Networks (CNN), particularly ResNets, have played a crucial role in improving accuracy in these advancements. Recent advancements include Vision Transformer (ViT) models, such as ViT-DR, which achieved 93.40% accuracy in classifying DR types using fundus images. Hybrid models combining CNNs with LSTMs, like DRNet23, have enhanced performance, reaching 96.50% accuracy. Multi-modal approaches leveraging different imaging techniques, such as the M3 deep learning framework, have also been developed to detect key DR. Dr. Lloyd M. Aiello [3] provides a comprehensive review of the evolution, current understanding, and treatment approaches for diabetic retinopathy. The author highlights the importance of key clinical trials that have influenced the management of DR, including its study and the early treatment demonstrating the effectiveness of laser photocoagulation. He advocates timely detection, evidence-based therapies, and comprehensive diabetes care to prevent DR-related vision loss. The 2021 review [5] outlines various glaucoma detection and management technologies, including imaging, functional testing, and emerging AI-based approaches. Optical Coherence Tomography and OCT angiography offer high-resolution imaging to identify retinal nerve fiber layer (RNFL) thinning and vascular abnormalities. Additional structural evaluations are supported by fundus photography and scanning laser polarimetry. The integration of AI in retinal imaging and teleophthalmology improves automated glaucoma screening, especially in areas with limited access to eye care. Artificial intelligence (AI)-driven retinal analysis and teleophthalmology enhance automated glaucoma screening, particularly in underserved regions. Jayaram et al. [6] offer an in-depth glaucoma analysis, highlighting the role of AI-based algorithms in evaluating ocular images, including color fundus photographs and optical coherence tomography (OCT) scans. These advancements support early detection and monitoring of glaucomatous changes, aiding healthcare professionals in diagnosis and treatment. A refined CNN was also designed explicitly to identify AMD from OCT scans [11]. The model is designed to balance efficiency and accuracy, making it well-suited for resource-constrained clinical settings. By employing architectural optimizations, the model achieves effective feature extraction from complex OCT data without excessive computational overhead. Abd El-Khalek et al. [14] propose a computer-aided diagnosis system designed for the early detection and classification of AMD. This system classifies fundus images into four categories: Geographic atrophy, intermediate AMD, wet AMD, and regular. It utilizes local and global feature extraction techniques to capture intricate retinal details by analyzing the entire retina and specific regions near the optical disc. The system employs a weighted majority voting approach to improve accuracy, integrating multiple classifier outputs for more reliable predictions. The model demonstrates high diagnostic performance, achieving an accuracy of 96.85 %, highlighting its effectiveness in AMD classification.

A comprehensive overview of diabetic retinopathy (DR), covering its epidemiology, risk factors, classification, and management, is presented in [2]. However, one major limitation is the lack of real-world clinical implementation data, including cost-effectiveness and accessibility in different healthcare settings. Dr. Lloyd M. Aiello's 2003 [3] paper on diabetic retinopathy lacks insights into AI-driven diagnostics, multimodal imaging, personalized medicine, telemedicine, anti-VEGF therapy, systemic biomarkers, and real-world data. Advancements in these areas highlight the need for updated research to enhance the screening, treatment, and management of diabetic retinopathy. The paper [5] thoroughly overviews glaucoma, covering risk factors, diagnostics, and treatment approaches. However, a few research gaps remain. Despite their growing role in early screening and disease progression prediction, the review does not extensively discuss AI-based diagnostic tools. Personalized medicine approaches, including genetic risk factors and pharmacogenomics, are also underexplored. While screening techniques are covered, the review does not discuss telemedicine's role in improving accessibility to glaucoma care. Jayaram et al. [6] provide an important roadmap for glaucoma management, but it lacks detailed insights into AI applications, precision medicine, neuroprotection, and telemedicine-based screening. Bridging these research gaps will improve early detection, treatment efficacy, and global accessibility to glaucoma care. Hsu et al. [11] present an efficient AI-driven approach for AMD detection, but several research gaps remain. The study lacks external validation across diverse datasets, limiting its generalizability to different populations and clinical settings. The CNN is a opaque model model with no explainability techniques (e.g., Grad-CAM) to help clinicians interpret its predictions.

It also does not incorporate longitudinal analysis of AMD progression, missing opportunities to track disease evolution over time. In [14], the model's generalizability is limited due to reliance on a single dataset, lacking multi-center validation across diverse populations and imaging conditions. The absence of explainability techniques (e.g., Grad-CAM, saliency maps) makes it difficult for clinicians to interpret AI decisions, hindering clinical adoption. Additionally, the study does not integrate multi-modal imaging data such as OCT or OCT-A, which could provide richer diagnostic insights.

After thoroughly reviewing the existing literature, we identified several research gaps that underscore the need for updated methodologies integrating AI-driven detection, multimodal imaging, personalized treatment approaches, and real-world implementation strategies to improve clinical outcomes. Future research should prioritize multi-center validation, hybrid deep learning architectures, and multi-modal data integration to enhance clinical adoption and real-world applicability. A key limitation of current AI-based disease identification models is their focus on single-disease classification, which restricts their broader applicability. To tackle this issue, the authors presented a multi-disease detection model to identify various ocular conditions, including DR, AMD, and Glaucoma. This approach broadens the application of AI in ophthalmic diagnostics. Additionally, existing disease classification methods predominantly rely on fundus images captured using specialized, high-cost imaging devices, such as fundus cameras, which are expensive, bulky, and require trained professionals for operation and maintenance. These limitations hinder their accessibility, particularly in remote or resource-constrained settings. To improve accessibility, affordability, and ease of use, we have developed a low-cost fundus imaging mechanism that is cost-effective, easy to assemble, and significantly more accessible than existing commercial options. This innovation aims to facilitate wider adoption of AI-based ophthalmic diagnostics, particularly in underserved regions, by reducing dependency on specialized imaging equipment and trained personnel.

To gain a comprehensive understanding of the technological landscape within our research domain, we explored a wide range of studies, not limited to ophthalmic diseases, but also various medical conditions. The objective was to analyse state-of-the-art methodologies in feature selection [15],enhanced image classification [16], and optimized image processing techniques. These studies provided valuable insights into technological advancements in medical imaging and pattern recognition.

Specifically, Ghazal et al. [17] and Siddiqui [29] were reviewed to investigate the integration of hardware and software components within intelligent medical systems. This study provided a framework for ensuring seamless interaction between various computational modules, enhancing system efficiency and reliability. Furthermore, Dhasarathan's [18] work was studied and examined to understand the best practices for safeguarding patient data, focusing on privacy-preserving techniques,secure data transmission,and regulatory compliance in healthcare AI applications.

These limitations highlight the urgent need for an integrated, scalable, cost-effective solution to address multiple eye diseases. By leveraging affordable hardware and advanced AI algorithms, a comprehensive system can improve early detection, reduce vision loss, and expand access to quality eye care. With an estimated 80% of global blindness being preventable or treatable, such a solution could significantly impact public health, especially in underserved regions.

## III. PROPOSED SOLUTION

Addressing the critical barriers of cost, accessibility, and usability, this system paves the way for democratizing eye care services. Early detection and timely intervention can significantly reduce the burden of vision loss, improving quality of life for millions worldwide. This holistic approach—leveraging cost-effective hardware, cutting-edge AI, and scalable cloud technology—significantly advances global eye health care, particularly for underserved communities. Fig. 1 illustrates the operation of our smart diabetic retinopathy system designed to detect comprehensive eye disease. It begins by taking images of the fundus using an IMX29 camera with a 28D lens set at an ideal distance to mimic the desired type of imagery. The images thus obtained are highly accurate and subject to processing, during which the essential features related to diabetic retinopathy and other eye conditions are extracted. Advanced classification is then carried out so the system provides diagnostic insights that help in timely intervention.

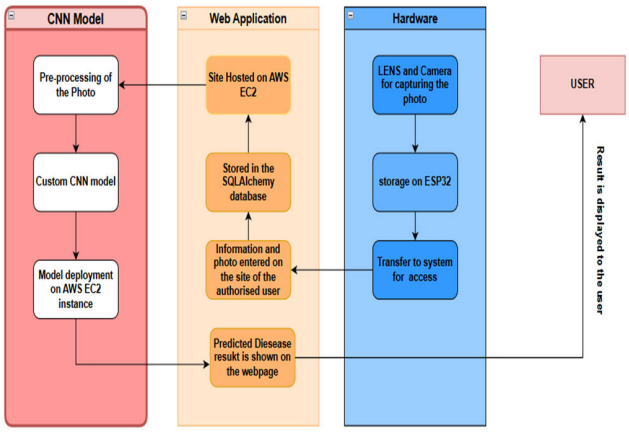



**FIGURE 1. Smart Diabetic Retinopathy System: From retinal image capture using an IMX29 camera to cloud-based AI diagnosis and user-friendly web access, the system offers an end-to-end solution for early and accurate eye disease detection.**

After capturing images, ESP32 can transfer captured images smoothly to the professional system and then upload them to the website, which is being run on the AWS cloud server for relay to analyze them further. Advanced AI models using CNN in the AWS setup accurately classify the disease in a fundus image. Through this integration of technology

and cloud-based infrastructure, our system thus offers precise diagnosis capabilities. Test results are shared on a user-friendly web app. The test results can thus easily be accessed from any internet-connected device, such as your phone or computer. The web app works like a helpful friend that walks you through all phases of your life with health-related matters.

Figure 10 lists every layer, kernel size, filter count, activation, output shape, and total parameters; Fig. 2 now mirrors this table. The architecture consists of the following components:

**Convolutional Blocks**: Four convolutional blocks with a convolutional layer, max pooling, and batch normalization extract and refine image features. Max pooling reduces spatial dimensions, and batch normalization stabilizes training.

Global Average Pooling: This layer compresses feature maps into a single vector, enhancing feature representation while minimizing the risk of overfitting.

Fully Connected Layers: The fully connected layers further process the extracted features, culminating in a final layer with softmax activation for multi-class classification.

Dropout Layers: To prevent overfitting, dropout layers are incorporated, randomly deactivating neurons during training. The model's architecture is designed to maintain a balance between computational efficiency and diagnostic precision, utilizing around 3.5 million parameters for effective learning. No layer uses pretrained weights; all model parameters were initialized from scratch using a random distribution suited for stable training with ReLU activations.

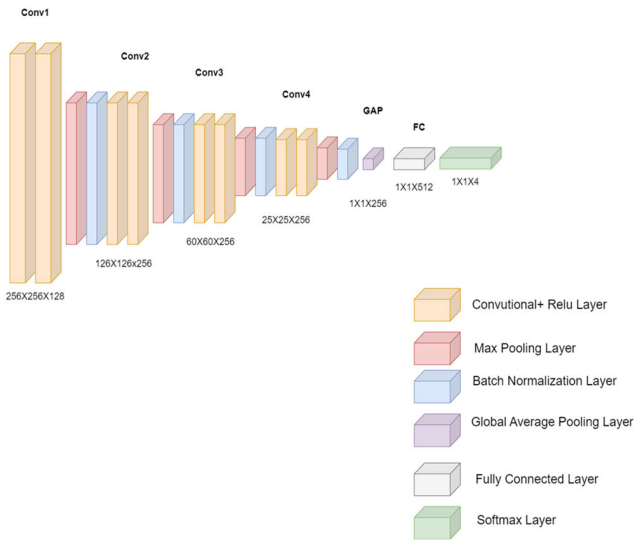



**FIGURE 2.** Proposed CNN model.

The key steps of the CNN network are as follows:

Step 1: Input Layer

The input layer consists of fundus images of people of diverse ages, Genders, and Areas. The inpur fundus images are of size (512,512).

Step 2: Convolutional Layer: The Convolutional layer forms the basis of the model. It can be expressed as given in equation 1 for this specific custom model consisting of multiple Convolution layers

$$g(x) = BN\,(MaxPool\,(Conv3\,(Conv3\,(x)))) \tag{1}$$

where g(x) represents the convolution function comprising of the Batch Normalization function (BN) and Max Pool represents the convolution layer is represented by C*onvi(x).*

Step 3: Relu Activation Function

The Rectified Linear Unit (ReLU) activation function is applied to the output of each convolutional layer to introduce non-linearity.

$$\text{ReLU} : A(x) = max\,(0, x) \tag{2}$$

where x represents the output received from the convolution function on which the relu function is applied.

Step 4: Max Pooling Layer: Max Pooling reduces the spatial dimensions by selecting the maximum value in a sliding window (e.g., $2 \times 2$). This helps in downsampling while retaining important features.The output determination is done as given in eqn. 3

$$y(i, j) = \max_{m,n \in R} x(i + m, j + n) \tag{3}$$

where $R$ is the pooling window (e.g., $2 \times 2$). m and n are the indices in it. $y(i,j) \rightarrow$ The output value at position $(i,j)$ after max pooling is the maximum value from the corresponding window in the input.

Step 5: Batch Normalization Layer: For each feature x in Batch Normalization normalizes activations across a mini-batch to stabilize training and speed up convergence. It rescales inputs using a learned mean and variance as given in eq. 4 and 5.

$$\hat{x} = \frac{x - \mu}{\sigma} \tag{4}$$

$$y = \gamma \hat{x} + \beta \tag{5}$$

where $\mu$ and $\sigma$ are batch mean and variance, and $\gamma, \beta$ are learnable parameters. $\hat{x}$ is the standardized version of input x, and y represents the scaled and shifted output. After normalization, the model learns two trainable parameters $\gamma$ (scale factor) $\rightarrow$ Adjusts the spread of the data. $\beta$ (shift factor) $\rightarrow$ Moves the data to a new mean. This allows the network to retain expressive power even after normalization

Step 6: Global Average Pooling Layer: Global Average Pooling (GAP) replaces fully connected layers by averaging each feature map as given in eqn. 6, reducing spatial dimensions to $1 \times 1$ while preserving depth.

$$y_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{i,j,c} \tag{6}$$

where H,W are height and width, and c is the channel index. $x_{i,j,c}$ The activation value at spatial position $(i,j)$ for channel $c$. $y_c \rightarrow$ The output for channel $c$ after applying Global Average Pooling.

Step 7: Fully Connected: After convolution and pooling, the feature maps are flattened and fed into fully connected

layers to combine the features learned by the previous layers using the eqn. 7.

$$f(x) = Dense(Dropout(Dense(GAP(g4(g3(g2(g1(x))))))))) \quad (7)$$

where $gi(x)$ represents each convolutional block. This results with final dimensions of $1 \times 1 \times 512$

Step 8: Softmax: A softmax layer is applied to the final output to produce a probability distribution over the classes (e.g., AMD, Glaucoma, DR). The softmax function can be calculated as shown in eqn, 8.

$$S(x) = \frac{e^{x_i}}{\sum_j e^{x_j}} \quad (8)$$

Step 9: Loss Function: Sparse Categorical Cross Entropy (calculated using eqn. 9) is the loss function used in this case. Used for multi-class classification with integer-labeled targets, it computes the negative log-likelihood of the correct class.

$$L = \frac{1}{N} \sum_{i=1}^{N} logp\left(\frac{y_i}{x_i}\right) \quad (9)$$

where L is the resultant loss function. $logp\left(\frac{y_i}{x_i}\right)$ is the predicted probability of the correct class for sample $i$.

Step 10: Adam Optimizer: Adam (Adaptive Moment Estimation) is an optimization algorithm combining Momentum and RMSprop, adapting learning rates per parameter.

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (10)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (11)$$

$$\theta_t = \theta_{t-1} - \frac{\alpha . m_t}{\sqrt{v_t} + \epsilon} \quad (12)$$

where $m_t$ is the First moment estimate (momentum). $g_t$ is Gradient of the loss function at time step t. $\beta 1$ is the Decay rate for past gradients. $v_t$ is the Second-moment estimate, whereas g sub t squared is the squared gradient at time step t. $\beta 2$ is the Decay rate for past squared gradients. $\theta_t$ is the updated model parameters. $\epsilon$ is a small constant to prevent division by zero.

## IV. EXPERIMENTAL DESIGN

Table 2 presents the system specifications used for experimental testing of proposed method.

The table provides demographic insights and dataset usage details for AI-based detection of the three diseases. The average patient age ranges from 60.6 years for Glaucoma to 67.4 years for AMD, with a slightly higher prevalence among females (55.5%–58.7%). The majority of patients are White (76.7%–81.2%) and Non-Hispanic (over 96%), with English being the primary language (∼91%). More than 50% of patients are married or partnered. Regarding diagnosis, 90% of DR cases are classified as non-vision-threatening, while most Glaucoma (51.3%) and AMD (64.3%) cases are labeled as usual. Mean deviation is recorded only for Glaucoma, ranging from 32.53 to 2.56. Across all three diseases, 60% of the dataset is used for training.

**TABLE 2. Experimental platform.**

| Entry No | System Elements | Minimum Specifications |
|---|---|---|
| 1 | Central Processing Unit | Intel Core i7 9700k |
| 2 | System Memory | 16 GB |
| 3 | Storage | 256 GB SSD |
| 4 | System software | Windows 10 |
| 5 | Coding Platform | Python |

### A. DATASET

Table 3 presents an overview of the dataset utilized for training, validation, and testing of the proposed model. Images originate from Harvard FairVision. We used stratified train/val/test splits of 64 / 16 / 20 %. Because the dataset is class-balanced, no re-weighting or synthetic sampling was applied. The training process employed the Adam Optimizer with a learning rate of 0.001, using categorical cross-entropy as the loss function. The model was trained for 50 epochs, incorporating an early stopping mechanism to mitigate overfitting. To assess performance, a stratified k-fold cross-validation approach was implemented, ensuring that each fold maintained the dataset's distribution. This strategy enabled a robust evaluation of the model's ability to generalize effectively.

### B. DATA PARTITIONING

The data is partitioned into three subsets so that the model trains and evaluates properly. For example, 60% of the data is assigned to train the model to learn the patterns and features. 20% of the data is kept for validation, fine-tuning the model, and avoiding overfitting by checking the performance during training. Finally, 20% is kept for testing so that the model is unbiased and properly generalized on unseen data.

The system was trained and validated using publicly available datasets, such as the ARIA dataset for AMD and the iChallenge-AMD dataset [28], which provide high-quality retinal images annotated with disease labels. These datasets were chosen for their extensive coverage of various retinal conditions, including diabetic retinopathy, glaucoma, and AMD.

Several preprocessing steps were applied to prepare the datasets for training. Fundus images were resized and cropped to ensure uniform dimensions, facilitating efficient model training. Normalization techniques were employed to standardize pixel intensity values, improving model convergence. Data augmentation techniques, such as rotation, flipping, and brightness adjustment, were applied to increase the diversity of the training data, reducing overfitting and enhancing the model's generalization capability.

**TABLE 3.** Details of the data set used for model training.

| Disease | Age | Gender | Top Race | Top Ethnicity | Top Language | Top Marital Status | Most Common Diagnosis | Mean Deviation | Dataset Usage (Training) |
|---|---|---|---|---|---|---|---|---|---|
| DR | 64.5 ± 16.5 | 55.5% Female 44.5% Male | White (77.8%) | Non-Hispanic (96.1%) | English (90.9%) | Married or partnered (54%) | Non-Vision threatening DR (90.0%) | - | 60% |
| Glaucoma | 60.6 ± 16.3 | 57% Female 43% Male | White (76.7%) | Non-Hispanic (96.2%) | English (92.6%) | Married or partnered (57.7%) | Normal (51.3%) | 4.26 (Range: 32.53 to 2.56) | 60% |
| AMD | 67.4 ± 17.1 | 58.7% Female 41.3% Male | White (81.2%) | Non-Hispanic (96.4%) | English (91%) | Married or partnered (52.6%) | Normal (64.3%) | - | 60% |

### C. HARDWARE AND SOFTWARE INTEGRATION

The fundus images were captured using a 28D lens paired with an IMX291 camera connected to an ESP32 microcontroller for image transmission. The processed images were uploaded to a web application hosted on AWS, where the trained DCNN model analyzed them for disease classification. This integrated system facilitated real-time diagnostics with high accuracy and user-friendly accessibility. By leveraging a combination of advanced AI techniques, robust datasets, and cloud-based infrastructure, the proposed method ensures reliable, scalable, and cost-effective diagnostics, addressing the critical challenges in the early detection of eye diseases.

## V. RESULTS AND DISCUSSION

To validate the proposed approach, three primary objectives are set to measure the effectiveness of the proposed model.

### A. PERFORMANCE METRIC USED

The metrics used include accuracy, sensitivity, specificity, precision, and F1-score, which assess the model's effectiveness. These metrics captured the system's ability to classify diseases while correctly minimizing false positives and negatives. Additionally, confusion matrices were analyzed to identify misclassification trends and areas for improvement. The accuracy of the experiment will be measured by its ability to correctly classify the occurrence of DR, AMD and glaucoma as defined below:

5.1.1 *Recall:* The degree of total True Positives (TP) or positive cases predicted as true is known as sensitivity. Also called as recall is given by eqn. 13

$$Recall = \frac{TP}{TP + FN} \tag{13}$$

where TP means cases where the defect is found, and FN depicts a defect found not actually there.

5.1.2 *Specificity*: Specificity is defined as the degree of actual negative value and is predictive as negative i.e.. True Negative (TN) means the case where no defect is found when it's not there. False Positive (FP) means the case when there is no defect, but the model predicts it. It can be calculated by using eqn. 14.

$$Specificity = \frac{TN}{FP + TN} \tag{14}$$

5.1.3 *Precision*: The percentage of actual results that are true or significant is known as precision and is given as shown in eqn. 15:

$$Precision = \frac{TP}{FP + TP} \tag{15}$$

5.1.4 *Accuracy*: Accuracy is defined as the percentage of Predicted results that are predicted correctly concerning actual values and is given in eqn. 16.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{16}$$

5.1.5 *F1 Score*: F1 score (F) presents a balanced model's of the Recall (R) and Precision (P). It can be calculated as given in eqn.17.

$$F1Score = \frac{2\ P.R}{P + R} \tag{17}$$

These equations form the backbone of the CNN model, enabling precise classification of fundus images for accurate diagnosis.

### B. WEB APPLICATION INTERFACE

Figure 3(c) Web App Patient Input Information page (part c) –(c) shows the Patient Input Data Page, this page is a data entry page where healthcare providers input patient-specific information, including, for instance, personal details, such as the patient's name, age, and medical history. A critical feature of this page is uploading retinal images captured by the IMX29 camera to predict diseases. The user interface makes image capture hassle-free and ensures that the format of images captures the critical features necessary to provide a definition diagnosis. The system asks for a re-shoot or re-upload if the image is wrong. Once validated, it processes the data for predicting diseases like diabetic retinopathy and glaucoma.

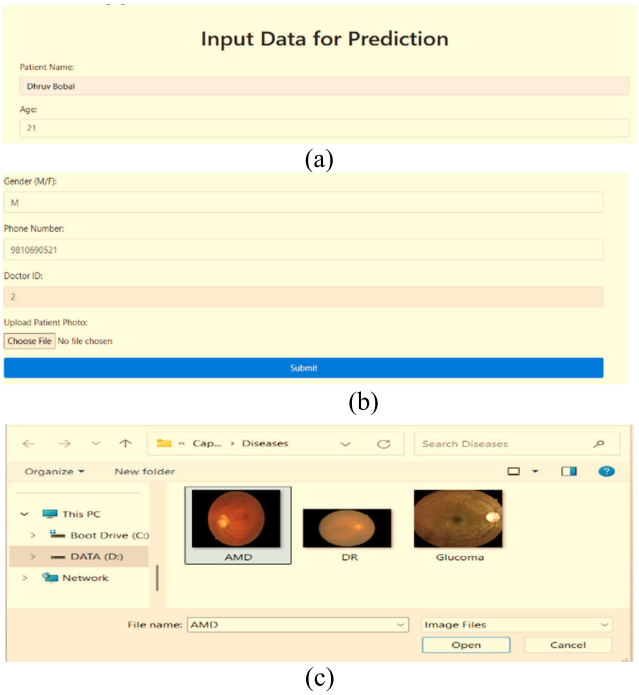



**FIGURE 3.** Web App Patient Input Information page1 (part a)(a), Web App Patient Input Information page1(part b) (b), Web App Patient Input Information page (part c) (c).

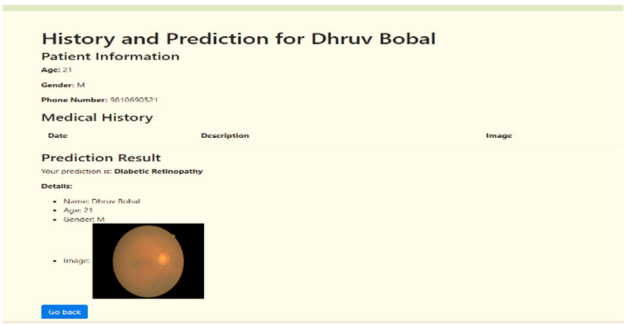



**FIGURE 4.** Patient result and history page of web app.

Figure 4 shows the Patient History and Result Sheet: The patient history and result page outline the individual's past diagnoses, including retinal imaging and predictive prognosis. For new additions, the page shows the latest results for predictions that outline the diagnosis with disease severity levels detected. Therefore, this section is of utmost importance in long-term patient management, as it allows clinicians to observe disease progress over time and make informed decisions based on past and present diagnostic information. It also ensures that continuous care is provided through a complete and readily available history of the patient's medical condition.

On examining the model accuracy graph illustrated in Fig 5, it appears that training accuracy, represented in red, starts at about 0.75 and boosts rapidly to a stable point around 0.95 after a few initial epochs, where it remains at this high level throughout the training. (The validation accuracy,

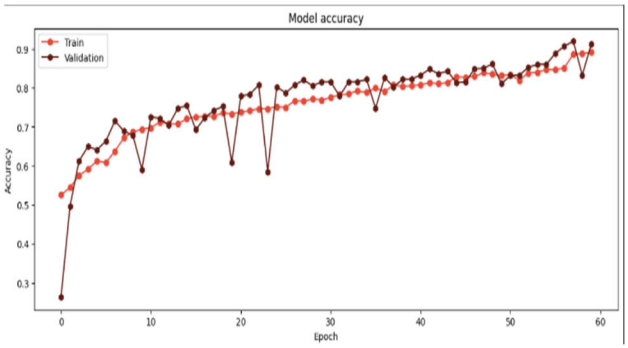



**FIGURE 5.** Model accuracy (showing epochs vs accuracy graph).

shown in brown, starts out lower around 0.25, but jumps very quickly in the first 5 epochs and is thereafter fairly oscillatory between about 0.8 and 0.9, showing a second apparent significant drop at epoch 20 where the accuracy temporarily falls to about 0.4, with an additional dip near epoch 30, suggesting some instability in the generalization performance of the model.).

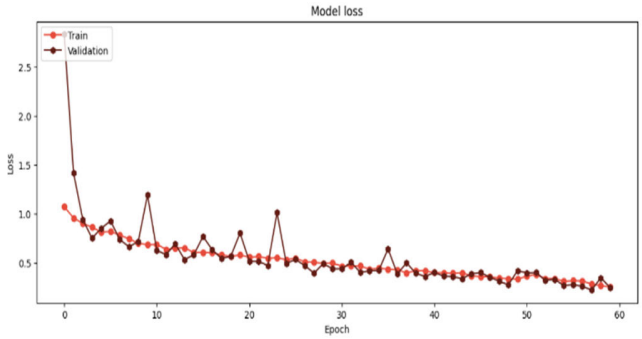



**FIGURE 6.** Model loss (showing epochs vs loss graph).

Examining the model loss graph in Fig 6, the training loss (shown in red) starts at around 0.8 and quickly decreases within the first few epochs, maintaining a consistently low loss value close to 0.2 throughout the training process. The validation loss, in brown, starts from a high value of about 3.5. It shows significant initial improvement during the first five epochs, after which it shows a critical volatility during the subsequent epochs. A sharp peak is seen at epoch 20, with the loss spiking to around 6.5, and another small peak near epoch 30. This characteristic may point to overfitting tendencies and suggests that the model can benefit from regularization or early stopping techniques.

Fluctuations observed in the validation loss during training indicate potential overfitting or instability in generalization. To address this in future iterations, the model can be refined through stronger regularization techniques such as incorporating L1 or L2 weight penalties. Additionally, the current dropout configuration may benefit from adjustment—both in rate and placement—by extending its use to earlier convolutional layers. Further improvements could also stem from experimenting with learning rate schedules, such as

cosine decay or cyclical policies, to encourage more stable convergence and smoother loss trajectories during training.

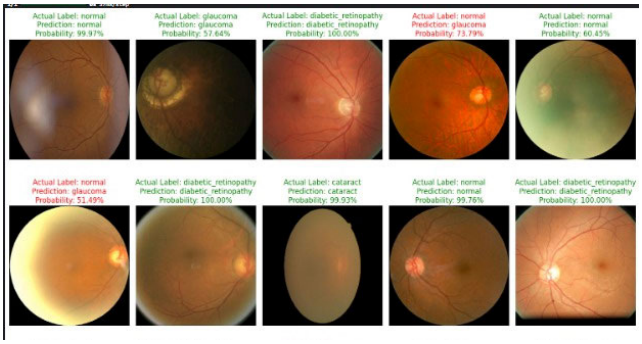


**FIGURE 7.** Sample Fundus images with model predictions (Part 1).

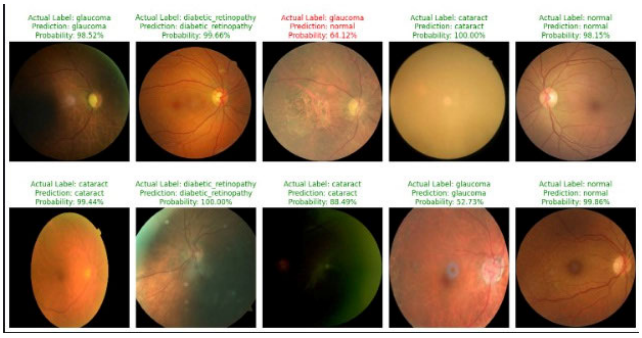


**FIGURE 8.** Sample Fundus images with model predictions (Part 2).

In Fig. 7-8, several retinal images, labels, and outputs from the trained machine learning model to screen against eye diseases are depicted. As such, the photos are arranged in grid format, and accompanying each image is the true diagnosis-for example, typical, diabetic retinopathy, cataract, glaucoma-the diagnosis predicted by the model and the probability associated with that prediction. The correct predictions are represented using green, while the wrong ones are represented by red.

For example, there have been numerous misclassifications where ''normal'' cases were classified as ''glaucoma,'' indicating that the model occasionally misinterprets. However, many images are well predicted with very high confidence levels, especially in diabetic retinopathy and cataracts, where probabilities reach almost 100%. The presence of misclassified cases, especially between normal and glaucoma cases, does, however, indicate potential avenues for improvement in the precision of the model. Overall, this visualization gives an overview of the model's performance, showing it as good in high-probability correct predictions but as vulnerable when handling more nuanced distinctions in images.

The confusion matrix shown in Fig 9 shows that the model performs well in all classes. In the case of cataracts, 194 instances were correctly classified, with a few wrong classifications as glaucoma and normal with 7 instances, respectively. Diabetic retinopathy has error-free predictions, with 220 correct predictions without a single mistake.

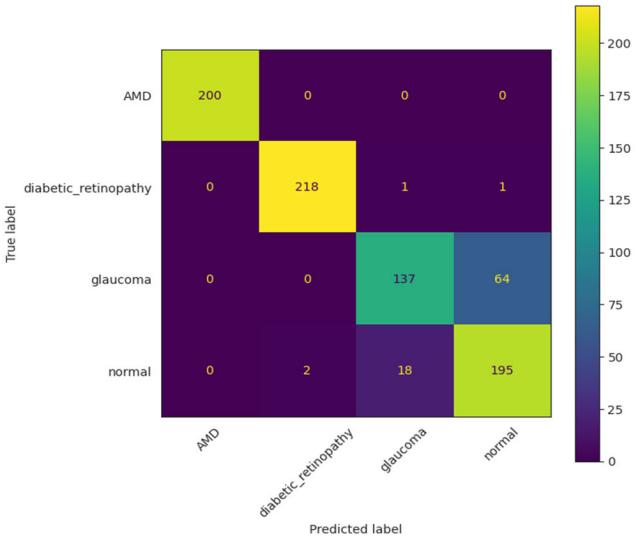



**FIGURE 9.** Confusion matrix for the model predictions.

Glaucoma has 165 correct predictions. However, there is some class ambiguity because 15 instances are classified incorrectly as cataracts and 21 as normal. The normal class has 191 correct predictions with some misclassifications, mainly for glaucoma (20 cases).

The training results in Fig. 10 indicate robust model performance across all evaluated datasets. On the training dataset, the model attained a loss of 0.16 and an accuracy of 98%, while the validation dataset exhibited a slightly higher loss of 0.14 with a comparable accuracy of 95%. Test set performance aligns with these validation results, at a loss of 0.18 and an accuracy of 96.5%, meaning the generalization is effective without significant overfitting.

The system's unique affordability, scalability, and diagnostic accuracy make it suitable for underserved regions. Its portability and cloud-based web application enhance usability, allowing for real-time feedback and timely medical interventions.

Figure 10 represents the model parameters and CNN architecture designed for image classification and consists of several Conv2D layers for feature extraction, MaxPooling2D layers for spatial down-sampling, and Batch Normalization layers for stabilizing training. The model has three main convolutional blocks, each followed by MaxPooling and Batch Normalization, allowing it to capture detailed features progressively. A GlobalAveragePooling2D layer condenses the feature maps, preparing them for the final Dense layers. The last Dense layer, with four units, corresponds to the four output classes. With around 3.5 million parameters, this model uses Dropout to reduce overfitting and improve generalization.

### C. COMPARISON OF MODEL PERFORMANCE METRICS

To compare the performance of our deep learning models, Custom CNN and previously proposed papers, we examine primary metrics (as given in eq. 13-17), such as accuracy,

| Layer (type) | Output Shape | Param # |
|---|---|---|
| conv2d (Conv2D) | (None, 256, 256, 128) | 3,584 |
| conv2d_1 (Conv2D) | (None, 256, 256, 128) | 147,584 |
| max_pooling2d (MaxPooling2D) | (None, 128, 128, 128) | 0 |
| batch_normalization (BatchNormalization) | (None, 128, 128, 128) | 512 |
| conv2d_2 (Conv2D) | (None, 126, 126, 256) | 295,168 |
| conv2d_3 (Conv2D) | (None, 124, 124, 256) | 590,080 |
| max_pooling2d_1 (MaxPooling2D) | (None, 62, 62, 256) | 0 |
| batch_normalization_1 (BatchNormalization) | (None, 62, 62, 256) | 1,024 |
| conv2d_4 (Conv2D) | (None, 60, 60, 256) | 590,080 |
| conv2d_5 (Conv2D) | (None, 58, 58, 256) | 590,080 |
| max_pooling2d_2 (MaxPooling2D) | (None, 29, 29, 256) | 0 |
| batch_normalization_2 (BatchNormalization) | (None, 29, 29, 256) | 1,024 |
| conv2d_6 (Conv2D) | (None, 27, 27, 256) | 590,080 |
| conv2d_7 (Conv2D) | (None, 25, 25, 256) | 590,080 |
| max_pooling2d_3 (MaxPooling2D) | (None, 12, 12, 256) | 0 |
| batch_normalization_3 (BatchNormalization) | (None, 12, 12, 256) | 1,024 |
| global_average_pooling2d (GlobalAveragePooling2D) | (None, 256) | 0 |
| dense (Dense) | (None, 512) | 131,584 |
| dropout (Dropout) | (None, 512) | 0 |
| dense_1 (Dense) | (None, 4) | 2,052 |

Total params: 3,533,956 (13.48 MB)
Trainable params: 3,532,164 (13.47 MB)
Non-trainable params: 1,792 (7.00 KB)

**FIGURE 10.** Model parameters and CNN architecture.

F1-score, specificity, and recall for all three disease classes: Diabetic Retinopathy, Glaucoma, and Age-related Macular Degeneration (AMD). The bar charts in Figures 11 to Figure 14 show a comparative analysis between the two models for each performance metric for the three disease classes.

The results of the proposed system demonstrate significant advancements compared to existing studies in eye disease detection. The achieved accuracy of 96.5%, along with high precision and F1-scores for diabetic retinopathy (98.2%), glaucoma (92.5%), and AMD (98.8%), surpasses many previous approaches that either focused on single-disease detection or required expensive hardware setups. For instance, studies leveraging convolutional neural networks (CNNs) for diabetic retinopathy detection often report 85% and 92% accuracy. Still, their models frequently struggle with dataset variability and lack multi-disease capabilities. The proposed system effectively addresses these limitations by integrating low-cost hardware and robust data augmentation.

The approach offers several strengths: affordability, scalability, and high diagnostic accuracy. Using a 28D lens and camera provides an accessible alternative to traditional fundus cameras, while advanced CNN architecture ensures the reliable detection of multiple diseases from a single framework. The cloud-based web application enhances usability and accessibility, particularly in resource-limited settings. Additionally, the integration of data augmentation improves model generalization, reducing overfitting and enhancing performance across diverse datasets.

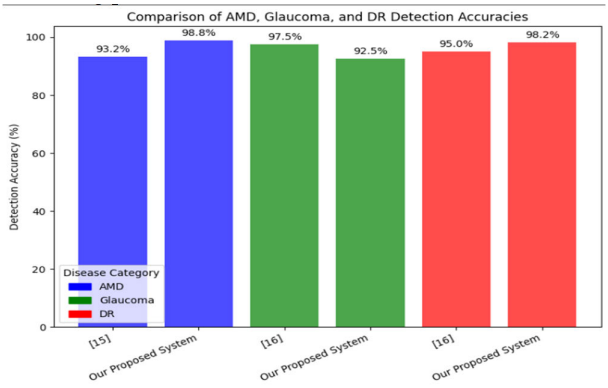



**FIGURE 11.** Comparison of AMD, Glaucoma and DR detection accuracies.

The graph presents a comparative analysis of AMD, Glaucoma, and DR detection accuracies across different models. Our proposed system demonstrates exceptional performance, achieving the highest accuracy in AMD (98.8%) and DR (98.2%), surpassing the best-performing alternatives. Although the MATLAB-Retrained AlexNet CNN slightly outperforms our model in Glaucoma detection (97.5% vs. 92.5%), the difference is minor.

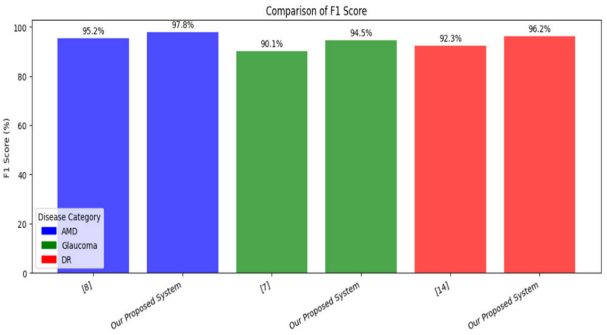



**FIGURE 12.** Comparison of AMD, Glaucoma, and DR detection F1 scores.

The proposed system demonstrates robust performance across multiple ocular conditions, achieving superior F1 scores compared to recent state-of-the-art approaches. Our system achieves an F1 score of 97.8% for AMD detection, showing a 2.6% improvement over AMDNet23's 95.2% [8]. In glaucoma detection, we attained an F1 score of 94.5%, outperforming Sharma et al.'s [7] score of 90.1% by 4.4%. For diabetic retinopathy detection, our system reached 96.2%, notably higher than Al-hazaimeh et al.'s [14] score of 92.3%. These consistent improvements across different ocular conditions demonstrate the robustness and versatility of our proposed approach.

The precision metrics (shown in Fig. 13) further validate our system's clinical reliability, with AMD detection achieving 97.5% precision compared to AMDNet23's [8] 94.8%, representing a 2.7% improvement in accurate positive predictions. For glaucoma, our system reached 93.9%

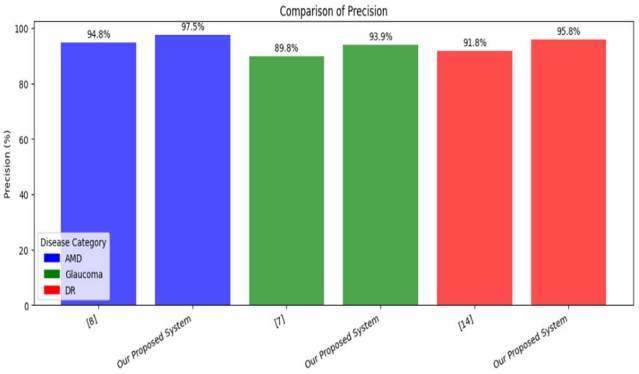



**FIGURE 13.** Comparison of AMD, Glaucoma and DR detection precision.

precision, surpassing Sharma et al.'s [7] 89.8% by 4.1%, while DR detection showed 95.8% precision, exceeding Al-hazaimeh et al.'s [14] 91.8% by 4%. These improvements in precision across all three conditions demonstrate the system's enhanced ability to minimize false positive diagnoses, which is crucial for maintaining clinical trust and preventing unnecessary interventions.

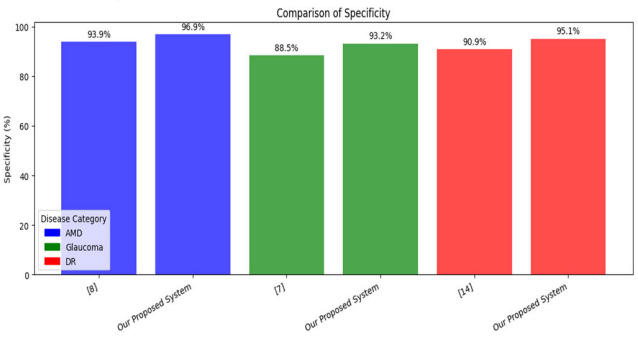



**FIGURE 14.** Comparison of AMD, Glaucoma, and DR detection specificity.

The specificity measurements in Fig. 14 showcase the system's exceptional ability to identify healthy cases, with AMD detection achieving 96.9% specificity versus AMDNet23's [8] 93.9%, showing a 3% improvement in accurately identifying non-diseased cases. In glaucoma detection, our system reached 93.2% specificity compared to Sharma et al.'s [7] 88.5%, while DR detection attained 95.1% against Al-hazaimeh et al.'s [14] 90.9%. These significant gains in specificity across all conditions underscore the system's reliability in correctly identifying healthy eyes, a critical factor for widespread screening applications where most cases may be normal.

Fig. 15 shows the comparative analysis of the Recall parameter of our proposed CNN against the previously proposed papers.

Figure 16 displays the Hardware structure developed for capturing Fundus images of the Eye. The Hardware consists of a 28D lens followed by an IMX291 Camera connected to ESP32, and the ESP32 is connected to a computer using a Micro USB cable for power connection.

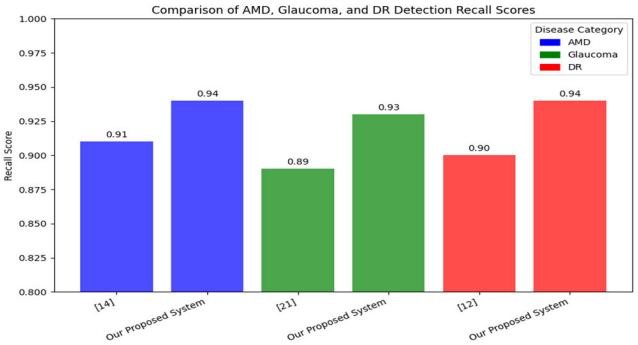



**FIGURE 15.** Comparison of AMD, Glaucoma, and DR detection recall.

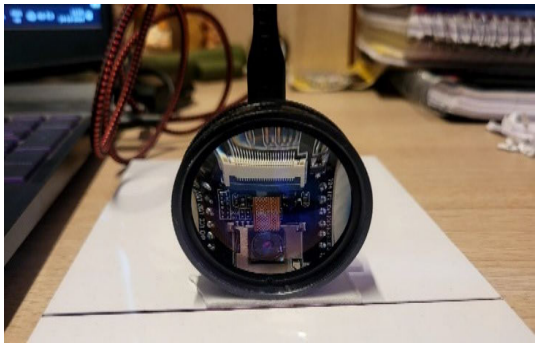


**FIGURE 16.** Hardware structure.

**TABLE 4.** Estimated budget.

| Component | Description | Approximate Cost (INR) |
|---|---|---|
| 28D Lens | magnification of the image | 4000 |
| Camera | High-resolution camera | 16000 |
| ESP32 | ESP32 with accessories | 480 |
| Bluetooth Module | Bluetooth Communication Module | 400 |
| Internet connectivity | 4G LTE USB modem or IoT SIM for data transmission | 1000 |
| Battery | Rechargeable drone battery and Raspberry Pi PSU | 6000 |
| Miscellaneous | Mounts, wiring, and hardware for integration | 5000 |

The approach offers several strengths: affordability, scalability, and high diagnostic accuracy. Using a 28D lens and camera provides an accessible alternative to traditional fundus cameras, while advanced CNN architecture ensures reliable detection of multiple diseases from a single framework. The cloud-based web application enhances usability and accessibility, particularly in resource-limited settings. Additionally, the integration of data augmentation improves model generalization, reducing overfitting and enhancing performance across diverse datasets. Table 4 presents the system's approximate budget. Unlike existing models specializing in detecting only a single disease, our approach provides a comprehensive, multiclass solution capable of

accurately detecting all three diseases. This eliminates the need for multiple models, making our system a more efficient and scalable alternative for real-world clinical applications.

However, certain limitations exist. Misclassifications observed in confusion matrices, particularly between normal and glaucoma cases, highlight the need for improved feature extraction and handling of subtle variations in optic nerve structures. Furthermore, while the system performs well with augmented datasets, real-world implementation may introduce challenges related to inconsistent image quality and lighting conditions. Further improvements may include refining the model to enhance sensitivity for borderline cases, particularly in glaucoma detection, and incorporating explainable AI techniques, such as Grad-CAM visualizations.

Overall, the proposed system represents a significant step forward in democratizing eye care, offering a scalable and cost-effective solution for the early detection of multiple diseases. By addressing current limitations and incorporating suggested improvements, the system has the potential to set new benchmarks in ophthalmic diagnostics, particularly in underserved regions.

## VI. CONCLUSION AND FUTURE WORK

The Eye Disease Detection System addresses the critical need for early and accurate diagnosis of severe eye conditions such as diabetic retinopathy, age-related macular degeneration (AMD), and glaucoma. By combining cost-effective hardware, advanced machine learning algorithms, and cloud computing, the system offers an accessible and scalable solution, particularly for communities with limited access to specialized eye care. Traditional diagnostic methods often require expensive equipment, such as fundus cameras, making early screening difficult for many. To overcome this barrier, the system employs a 28D lens and an ESP32 microcontroller to capture high-quality retinal images at a significantly lower cost. These images are analyzed using an AI-powered convolutional neural network (CNN) model, achieving an impressive 95.5 % accuracy while enabling multi-disease detection. A dedicated web application further enhances accessibility by providing healthcare professionals with a user-friendly interface, reducing diagnostic delays, and offering real-time feedback. However, despite its high accuracy, the system faces challenges such as misclassifications, inconsistent image quality, and the opaque decision-making process of CNN models. Future advancements will focus on refining the model to improve accuracy in borderline cases, incorporating explainable AI techniques like Grad-CAM for greater transparency, expanding datasets to enhance robustness across diverse populations, integrating telemedicine for remote diagnostics, and extending capabilities to detect additional conditions such as cataracts and retinal vein occlusion. By bridging the gap between affordability and advanced technology, this system has the potential to revolutionize ophthalmic diagnostics. Continuous improvements can democratize eye care, making early disease detection accessible to underserved communities worldwide.

## ACKNOWLEDGMENT

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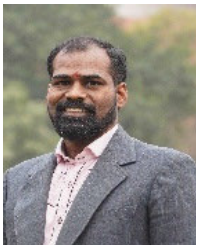
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